

ABSTRACT This project presents an intelligent document analysis system, the system combines OCR and LLM to handle the PDF files and images. On this way, it improve the reliability to understanding the document rather than only focus on the text extraction. Then the system will evaluate the text quality and detect the structure of the document. Base on these results, system will decide whether the text is suitable or not. This helps to avoid low quality OCR results or indescribable types of document such as charts. After finish the preprocessing, the system will use LLM to generate structured outputs including Summary, Key Data, Important Terms, and Reading Recommendations. The system also provides a UI interface for users to upload the files, view the results and ask questions. This reference the ideas from RVL-CDIP and FUNSD. In general, the system shows a complete and practical way to analysis the document, the key point is about quality control and structure sensitive processing

In recent years, more and more documents have been stored as digital files such as PDF and images. These documents often contain useful information but its not easy to extract and understand the content automatically. Traditional OCR systems can convert images into text, but they often fail when the image quality is low or when the document structure is bit complex. At the same time, the large language models can help analyze text and generate summaries. However, these models depend on the quality of the input text. If the OCR result is poor, the final analysis will also be unreliable. In this project, we design an intelligent document analysis system that combines OCR and language models. The system can process both PDF files and images. It first extracts text using OCR, then checks the quality of the extracted text and finally performs structured analysis. The main goal of this system is not only to extract text, but also to improve reliability. To achieve this, we introduce OCR quality evaluation and document structure detection. These will help the system decide whether the extracted text is suitable for further analysis. The OCR quality evaluation is based on simple but effective rules. The system considers factors such as text length, the ratio of valid characters, and the presence of abnormal symbols. If the extracted text contains too many irregular patterns or if is too short, the quality score will be low. This rule based approach is simple and easy to apply, at the same time it can effectively filter out the unreliable OCR output results. The system also provides a UI for user to upload any files, checking analysis results and ask any questions. This makes the system more easier for users to use.